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<http://lukaseigentler.github.io>

Variation in resource quality causes maintenance of individual variation in a competitive trait

BIRS workshop on nonlocal aggregation models

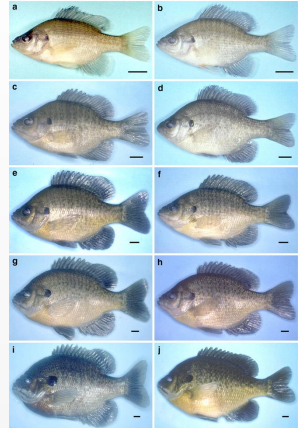
24 August 2026

Lukas Eigentler (University of Warwick)
joint work with

Klaus Reinhold (Bielefeld), David W. Kikuchi (Oregon State), Valeria Giunta (Swansea)

Individual variation

- Evolution does not necessarily lead to identical individuals.
- Individual variation is ubiquitous in many populations.
- This includes **genetic variation**.
- Potential causes: temporal/spatial environmental variability, negative frequency-dependent selection, ...



Bluegill size differences¹

¹ (Yokogawa, K.: *Ichthyological Research* 60.1 [2012])

Individual variation in competitive traits

- Traits determining intraspecific competitiveness often vary within populations.
- Examples: armaments, body size, sensory capabilities to detect food, ...
- **Investment into competitiveness typically bears a cost.**
- Trade off between ability to attain resources and resource investment into reproduction.



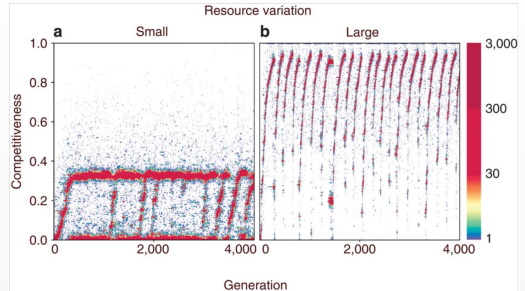
Source: Wikimedia Commons

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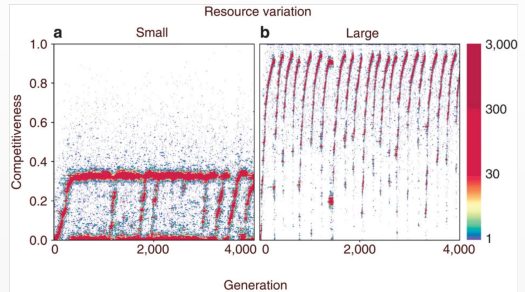


Source: Wikimedia Commons

(Baldauf, S. A., Engqvist, L. and Weissing, F. J.: *Nat Commun* 5.1 [2014])

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- What if resource qualities also vary continuously?



Source: Wikimedia Commons

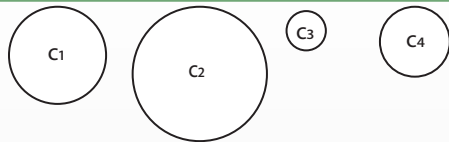
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Individual based model



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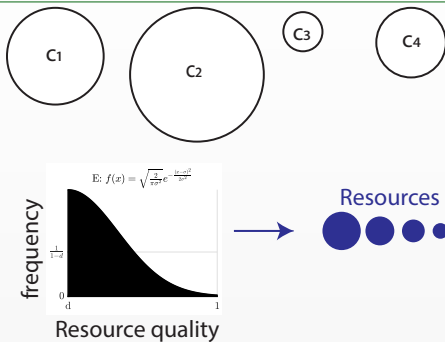
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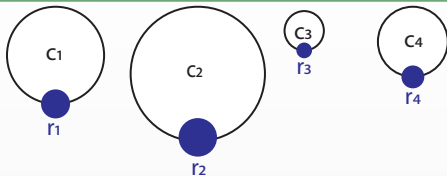
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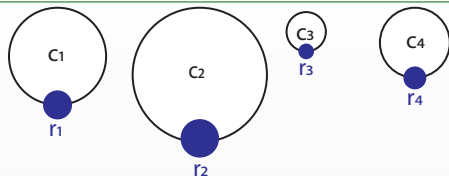
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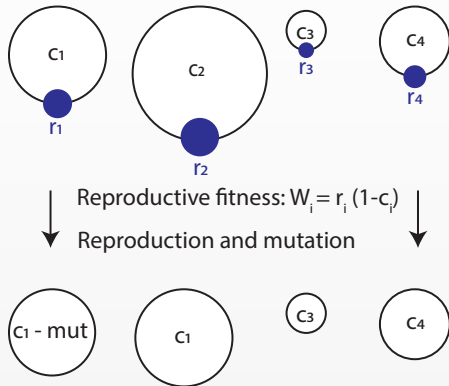
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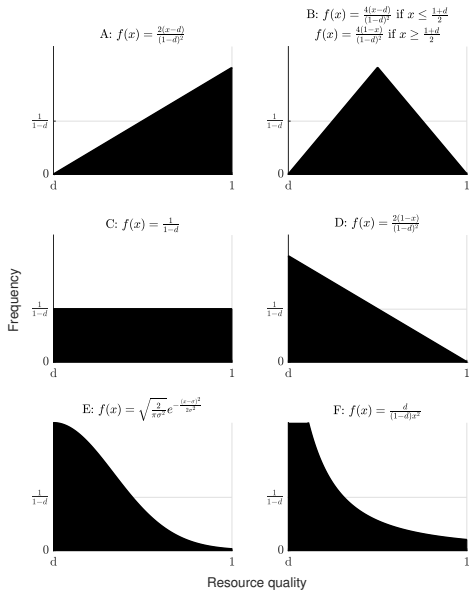
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- Reproductive fitness is $W_i = r_i(1 - c_i)$.
- Next generation is created through draw with replacements with weights W_i and mutations.



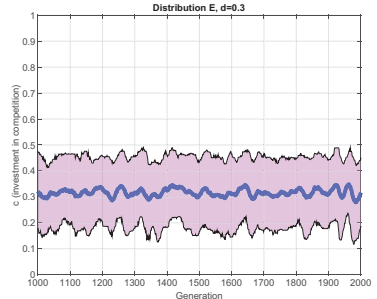
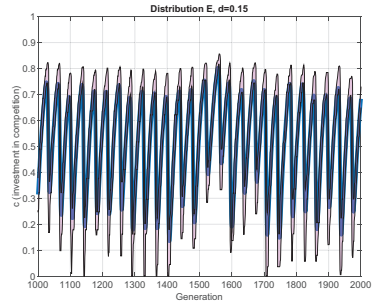
Resource distributions

- A: linearly increasing
- B: linearly increasing and decreasing
- C: uniform
- D: linearly decreasing
- E: half-normal
- F: power law



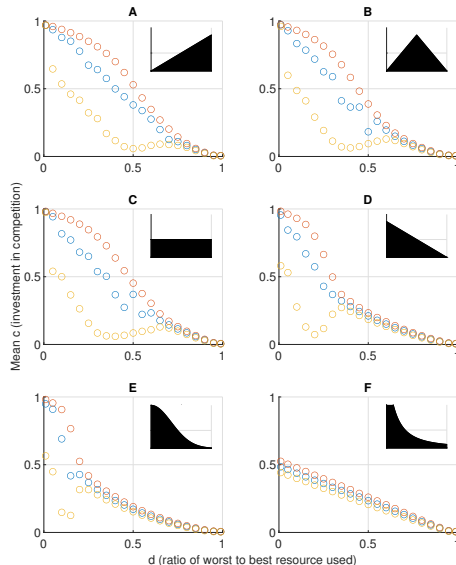
Arms races vs. stable polymorphisms

- Dynamics either represent **arms races** or **stable polymorphisms**.
- Arms race: mean competitive trait oscillates with little individual variation at a time.
If all individuals have $c = c^*$, then $c = c^* + \varepsilon > c^*$ has higher fitness.
If c^* is high, low competitive traits can invade; in particular, $c_1 = 0$ (fitness $W = (1 - c_1)d = d$) has larger fitness than $c_2 > 1 - d$ because $W = (1 - c_2)r < dr \leq d$.
- Stable polymorphism: mean competitive trait remains constant over time; large individual variation at all times.



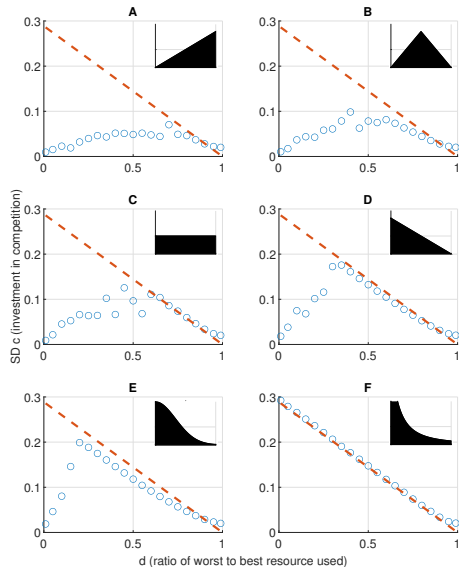
Worst-to-best resource ratio

- The ratio d of the worst to the best resource determines mean investment into competition.
- Arms races occur when d is small.
- Period of arms race cycles increases as d decreases; unclear behaviour for $d \rightarrow 0$.
- No arms races for power law distribution.
- Stable mean trait values occur when d is large.
- The threshold $d = d^*$ depends on the resource distribution.



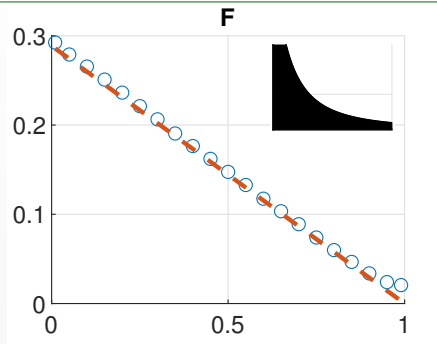
Worst-to-best resource ratio

- Little individual variation occurs during arms races.
- Stable mean trait values are always associated with stable polymorphisms.
- Individual variation is close to maximum (uniform distribution with $0 \leq c \leq 1 - d$).



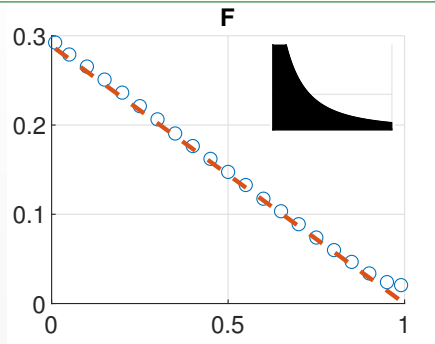
Special case: power law

- Can show analytically that the power law with pdf $g(r) = \frac{d}{(1-d)r^2}$ leads to uniform trait distribution.



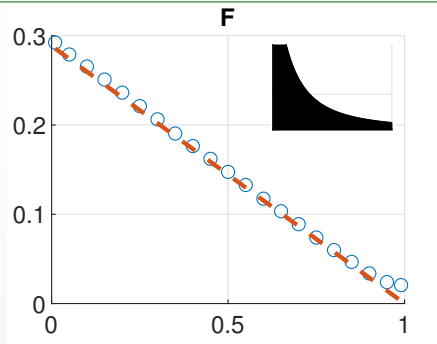
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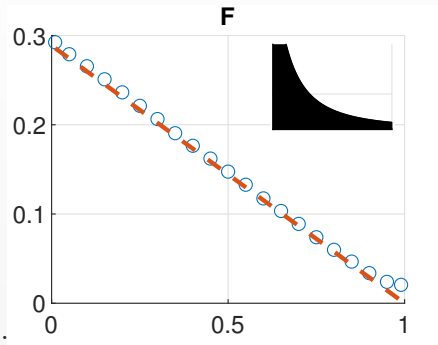


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Therefore, the cdf of the trait distribution is

$$P(c < c^*) = P(r < r^*) = \frac{r^* - d}{(1 - d)r^*} = \frac{c^*}{1 - d}.$$

$\Rightarrow$ uniform distribution.

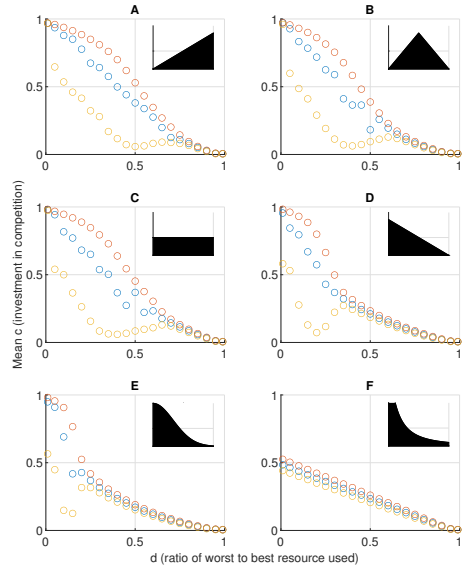


Conclusions – IBM

- Continuous individual variation in a competitive trait is a consequence of intraspecific competition for resources with continuously varying quality.
- The ratio between worst and best quality resources determines (i) mean investment into competition, and (ii) whether arms races with little individual variation or stable polymorphisms occur.
- The distribution of resource quality is important.
- Resource distributions with more low-quality resources lead to stable polymorphisms in more cases.

What next? Nonlocal PDE!

- Main limitation: IBMs are hard to analyse.
- PDE models allow for bifurcation analysis.
- Formal derivation of a PDE from the IBM is (we think) not possible.



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- $$\frac{\partial u(t, c)}{\partial t} = m \frac{\partial^2 u(t, c)}{\partial c^2} + \lambda(t, c)u(t, c).$$

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- Average fitness

$$\overline{W}(t) = \int_0^\infty W(t, c) u(t, c) dc .$$

- Fitness of phenotype c

$$W(t, c) = \underbrace{(1 - c)}_{\text{proportion left for reproduction}} \cdot \underbrace{r(t, c)}_{\text{resource quality attained}} .$$

Resource quality

- For $0 \leq d < 1$, let the pdf of the resource distribution be $g : [d, 1] \rightarrow [0, \infty)$, such that the inverse $G^{-1} : [0, 1] \rightarrow [d, 1]$ of its cdf $G : [d, 1] \rightarrow [0, 1]$ exists.

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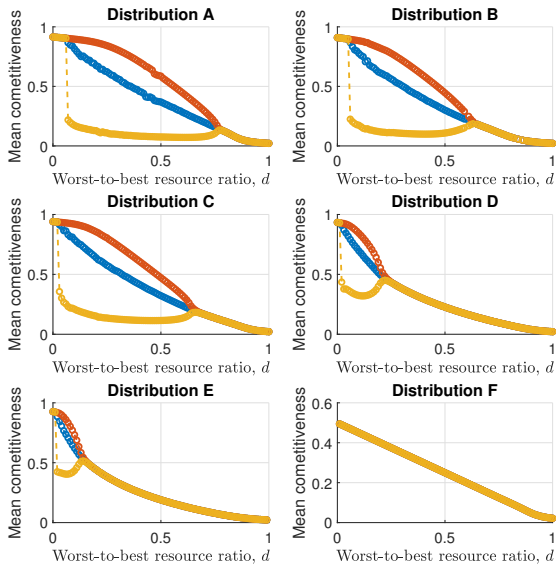
$$\Rightarrow r(t, c) = G^{-1}(U(t, c)).$$

Full nonlocal model

$$\frac{\partial u(t, c)}{\partial t} = m \frac{\partial^2 u(t, c)}{\partial c^2} + \left[(1 - c) G^{-1} \left(\int_0^c u(t, \tilde{c}) d\tilde{c} \right) - \int_0^\infty (1 - c) G^{-1} \left(\int_0^c u(t, \tilde{c}) d\tilde{c} \right) u(t, c) dc \right] u(t, c).$$

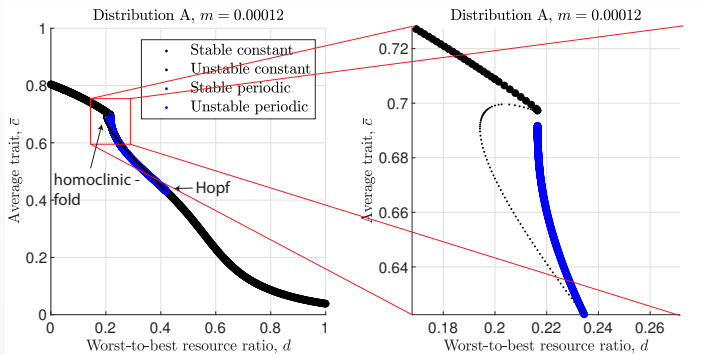
Numerical simulations

- Results qualitatively the same as for IBM.
- Note lack of oscillations for small d .



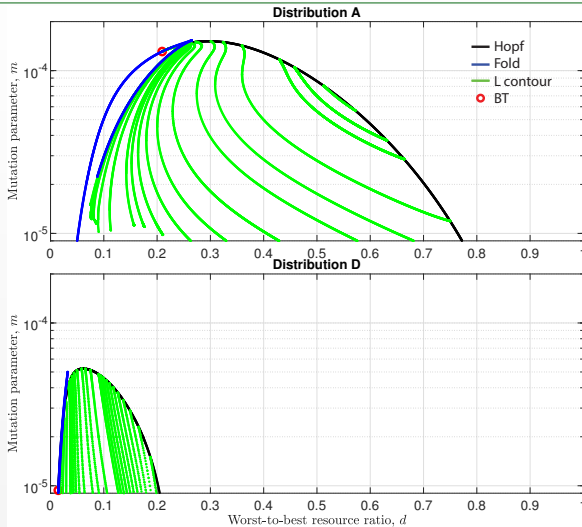
1d numerical bifurcation analysis **work in progress**

Tool: discretise trait space $\Rightarrow$ continuation of large ODE system using AUTO-07p.



From high d to low d : constant sol $\rightarrow$ onset of periodic sol via Hopf bifurcation $\rightarrow$ termination of periodic sol at either Hopf bifurcation or homoclinic-fold bifurcation $\rightarrow$ constant sol.

2d numerical bifurcation analysis: m - d **work in progress**



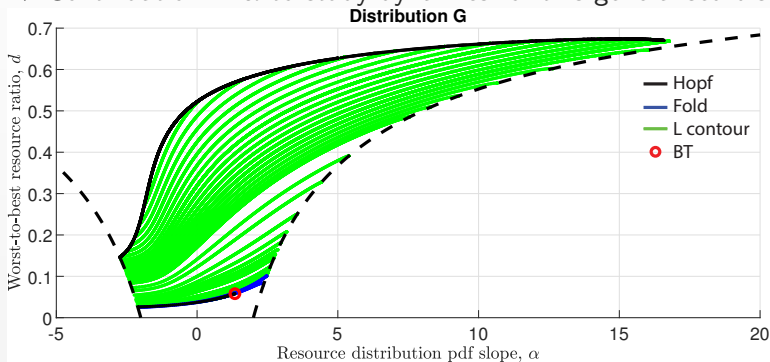
Periodic sol require sufficiently low mutation rate. Onset at low d occurs at Hopf bifurcation for “large” m ; at homoclinic-fold bifurcation at “small” m .

2d bif analysis: generalised linear distribution **work in progress**

Distributions A, D are special cases of general distribution with pdf

$$g(r; \alpha) = \alpha r + \beta, \quad d \leq r \leq 1, \quad -\frac{2}{(1-d)^2} \leq \alpha \leq \frac{2}{(1-d)^2}$$

⇒ Continuation in α to study dynamics for this generalised distribution.



Analytical results **work in progress** – Valeria Giunta

We can show analytically that

- the existence of lower and upper bounds for steady state solutions.
- if the fitness is monotonically decreasing with the trait, the trait distributions are monomodal with peak at $c = 0$.
- there are sufficient conditions for constant solutions to be stable.

Conclusions – PDE

- Numerical bifurcation analysis of PDE model is possible.
- Pattern onset at Hopf bifurcation (high/low d) and homoclinic-fold bifurcation (low d).
- Can perform 2-parameter continuation of these bifurcations.
- Low mutation rates required for periodic “arms race” solutions.
- Can find codimension 2 bifurcation, such as Bogdanov-Takens.
- Can perform continuation in a parameter describing the resource quality distribution to understand how different distributions affect results.

Open questions

- What are the most realistic resource quality distributions?
- How to include more realism, such as spatial dynamics or sexual reproduction?
- Are there better tools for numerical bifurcation analysis of nonlocal PDEs (BifurcationKit, pde2path)?
- Can we formally derive a continuum model from the IBM?

References

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- [1] Reinhold, K., Eigentler, L. and Kikuchi, D. W.: 'Evolution of individual variation in a competitive trait: a theoretical analysis'. *J. Evolution. Biol.* 37.5 (2024), pp. 538–547.